The Perron–Frobenius Rank-One Step in Cirac–Perez-Garcia–Schuch–Verstraete 2017 Appendix C.2

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1 The assertion

Appendix C.2 of Cirac–Pérez-García–Schuch–Verstraete relates the strong area law and zero correlation length for a simple matrix product density operator to an outer-product factorization of a finite trace matrix [CPGSV16, Appendix C.2, Lemmas C.3–C.4].[1]

The paper constructs positive neighboring operators $\eta_{k,h}$ and defines

$$T_{k,h} = \mathrm{tr}(\eta_{k,h}).$$

Thus T is entrywise non-negative. Under the strong area law the source argues that T is primitive, in the sense that some power of T has all entries strictly positive. Under zero correlation length it obtains

$$\mathrm{tr}(T^N) = \mathrm{tr}(T)$$

for every positive integer N, with $\mathrm{tr}(T) = 1$ after normalization. The proof then invokes fixed-point theory for primitive non-negative matrices and concludes that T has one eigenvalue equal to 1 and all other eigenvalues equal to 0. From this it concludes the desired factorization

$$T_{k,h} = a_k b_h.$$

The last implication is the point at issue. Primitivity and the equalities of positive-power traces control the nonzero eigenvalues of T, but they do not exclude nilpotent Jordan structure on the generalized eigenspace for the eigenvalue 0. Such nilpotent structure is invisible to traces of positive powers and can prevent T from having rank one.

[1] For traceability, this note corresponds to issue #1041. The underlying question was isolated in issue #832; the finite counterexample was added in pull request #849; and the positive-semidefinite replacement theorem was added in pull request #917. The source passage is `Papers/1606.00608/MPDO-22-12-17-2.tex:1394--1499`.

2 The counterexample

Consider the real 3×3 matrix

$$T = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

All entries are non-negative, and

$$T^2 = P := \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Hence T is primitive, since T^2 has all entries strictly positive. Also $P^2 = P$ and $PT = TP = P$, so $T^N = P$ for every $N \geq 2$. Since $\mathrm{tr}(T) = \mathrm{tr}(P) = 1$, it follows that

$$\mathrm{tr}(T^N) = \mathrm{tr}(T) = 1$$

for every positive integer N.

Nevertheless T is not an outer product. If $T = ab^\top$, then the entry $T_{1,1} = 1/2$ implies $a_1 \neq 0$. The entry $T_{1,3} = 0$ then forces $b_3 = 0$, but this contradicts $T_{2,3} = 2/3$. Thus the universal matrix implication

$$\text{primitive} + \text{constant positive-power traces} \Longrightarrow \text{rank one}$$

is false for finite entrywise non-negative real matrices. [2]

This example has the form $T = P + \frac{1}{6}xy^\top$, where $x = (1, -1, 0)^\top$ and $y = (1, 1, -2)^\top$. The perturbation satisfies $(xy^\top)^2 = 0$ and is annihilated by P on both sides. It therefore leaves all traces of positive powers unchanged from the Perron projector while raising the rank of T.

3 The corrected statement

The formal development proves a corrected sufficient matrix theorem. Let T be a finite real square matrix. If T is positive semidefinite, $\mathrm{tr}(T) = 1$, and $\mathrm{tr}(T^N) = \mathrm{tr}(T)$ for every positive integer N, then T has an outer-product factorization $T = ab^\top$. [3] Primitivity is not needed for this corrected matrix theorem once positive semidefiniteness and trace normalization are assumed.

The proof is spectral. Positive semidefiniteness makes T Hermitian and diagonalizable with non-negative real eigenvalues λ_i. The trace normalization gives $\sum_i \lambda_i = 1$. The trace-power hypothesis at $N = 2$ gives $\sum_i \lambda_i^2 = 1$. For

[2] The formal counterexample is `TNLean/Archive/PerronFrobeniusRankOneCounterexample.lean`, especially the matrix definition at lines 63–65 and the theorem `counterexample` at lines 150–154. It is also summarized in `docs/counterexamples.md:19--27`.

[3] The formal statements are `Matrix.PosSemidef.trace_powers_constant_implies_rank_one` and `Matrix.primitive_trace_powers_constant_implies_rank_one_of_posSemidef` in `TNLean/Algebra/PerronFrobenius/RankOne.lean:181--240`.

non-negative numbers with sum one, this forces exactly one λ_i to be 1 and all the others to be 0. Therefore the diagonal form has rank one, and unitary conjugation gives an outer-product factorization of T.

This corrected theorem does not refute the intended MPDO result. It identifies a matrix-level hypothesis sufficient for the rank-one step. The remaining question is whether the trace matrix $T_{k,h} = \mathrm{tr}(\eta_{k,h})$ arising from the neighboring operators is Hermitian or positive semidefinite. Positivity of each $\eta_{k,h}$ gives $T_{k,h} = \mathrm{tr}(\eta_{k,h}) \geq 0$ entry by entry, but entrywise nonnegativity is not matrix-level positive semidefiniteness or Hermitian symmetry of T. [4]

4 Conclusion

The proof in Appendix C.2 contains an unjustified matrix step as written: a primitive entrywise non-negative matrix with constant traces of positive powers need not have rank one. The obstruction is a nilpotent Jordan component at the zero eigenvalue, which is not detected by the trace identities.

The formal development proves a corrected sufficient theorem: positive semidefiniteness, trace normalization, and the same trace-power hypothesis imply the required rank-one factorization. What remains for the unconditional MPDO argument is not another Perron–Frobenius statement for arbitrary nonnegative matrices, but an MPDO-specific proof that the trace matrix obtained from the η-operators has the Hermitian or positive-semidefinite structure needed by the corrected theorem. Thus the source proof has a genuine unsupported intermediate assertion, while the intended MPDO theorem itself has not been disproved.

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.

[4] The entrywise nonnegativity result is `MPOTensor.ExplicitEtaOperators.traceMatrixRe_nonneg` in `TNLean/MPS/MPDO/SimpleLocalStructure.lean:306--318`. The positive-semidefinite reformulation for the rank-one conclusion is `MPOTensor.sal_zcl_implies_rank_one_T_of_posSemidef` in `TNLean/MPS/MPDO/SimpleLocalStructure.lean:350--366`. The corresponding blueprint discussion is `blueprint/src/chapter/ch02b_mpdo.tex:568--680`.